

◆ PHASE 1: Python Foundation (Weeks 1–6)

✓ Week 1 (Sunday Target)

- Python basics
- Variables, data types
- Input/output
- If-else
- Mini programs

🎯 Sunday Goal:

→ Write small Python programs **without help**

✓ Week 2

- Loops (for, while)
- Functions
- Code structure
- Debugging basics

🎯 Sunday Goal:

→ Solve 10–15 logic problems in Python

✓ Week 3

- Lists, tuples, sets
- Dictionaries
- String manipulation

🎯 Sunday Goal:

→ Comfortably use collections & loops together

✓ Week 4

- OOP in Python
- Classes & objects
- File handling

🎯 Sunday Goal:

→ Build a small console-based project

✓ Week 5

- NumPy basics
- Arrays
- Matrix operations

🎯 Sunday Goal:

→ Understand arrays & vector operations

✓ Week 6

- Pandas
- CSV handling
- Data cleaning
- Basic visualization

🎯 Sunday Goal:

→ Load, clean & visualize a dataset

◆ PHASE 2: Math for AI (Weeks 7–10)

✓ Week 7 – Statistics

- Mean, median, variance
- Standard deviation
- Probability basics

🎯 Sunday Goal:

→ Understand data distribution

✓ Week 8 – Probability

- Conditional probability
- Bayes theorem
- Normal distribution

🎯 Sunday Goal:

→ Understand classification intuition

✓ Week 9 – Linear Algebra

- Vectors
- Matrices
- Dot product
- Matrix multiplication

🎯 Sunday Goal:

→ Understand how data flows in ML

✓ Week 10 – Calculus

- Derivatives
- Gradients
- Chain rule (intuition only)

🎯 Sunday Goal:

→ Understand **how models learn**

◆ PHASE 3: Machine Learning (Weeks 11–15)

✓ Week 11

- What is ML?
- Supervised vs Unsupervised
- ML workflow

🎯 Sunday Goal:

→ Understand ML pipeline clearly

✓ Week 12

- Linear Regression
- Cost functions
- Gradient descent

🎯 Sunday Goal:

→ Build your first ML model

✓ Week 13

- Logistic Regression
- Classification metrics
- Confusion matrix

🎯 Sunday Goal:

→ Build a classifier model

✓ Week 14

- Decision Trees
- Random Forest
- Overfitting & underfitting

🎯 Sunday Goal:

→ Understand model selection

✓ Week 15

- KNN, SVM, Model evaluation

🎯 Sunday Goal:

→ Compare multiple ML models

◆ PHASE 4: Deep Learning (Weeks 16–18)

✓ Week 16

- Neural networks
- Activation functions
- Loss functions

🎯 Sunday Goal:
→ Understand neural network basics

✓ Week 17

- Backpropagation
- Optimizers
- CNN basics

🎯 Sunday Goal:
→ Train a simple neural network

✓ Week 18

- PyTorch / TensorFlow
- Image classifier project

🎯 Sunday Goal:
→ Build a deep learning project

◆ PHASE 5: Generative AI + Projects (Weeks 19–22)

✓ Week 19

- What is Generative AI
- Transformers (high-level)
- Attention mechanism

🎯 Sunday Goal:
→ Understand ChatGPT-like models

◆ PHASE 5: Generative AI + Projects (Weeks 19–22)

✓ Week 19

- What is Generative AI
- Transformers (high-level)
- Attention mechanism

🎯 Sunday Goal:
→ Understand ChatGPT-like models

✓ Week 20

- Hugging Face
- Prompt engineering
- Using pretrained models

🎯 Sunday Goal:
→ Build an AI chatbot

✓ Week 21

- Major AI project
- GitHub documentation
- Portfolio integration

🎯 Sunday Goal:
→ One **industry-ready AI project**

✓ Week 22 (Before June 4)

- Revision
- Resume update
- Internship applications
- Interview prep basics

🎯 Sunday Goal:
→ **AI-ready by June 4** 🎉

Weekly Study Pattern (Recommended)

- Mon–Fri: 2–3 hrs/day
- Saturday: Practice + projects
- Sunday:
 - ☒ Revise
 - ☒ Check weekly goal
 - ☒ Plan next week